But in the 1970s, Jacob Bekenstein and Stephen Hawking discovered that this isn't right. Their mathematical analyses showed that the entropy of a black hole is not proportional to its volume, but instead is proportional to the *area* of its event horizon—roughly speaking, to its surface area. This is a very different answer. Were you to double the radius of a black hole, its volume would increase by a factor of 8 (2^3) while its surface area would increase by only a factor of 4 (2^2); were you to increase its radius by a factor of a hundred, its volume would increase by a factor of a million (100^3), while its surface area would increase only by a factor of 10,000 (100^2). Big black holes have much more volume than they do surface area.[2] Thus, even though black holes contain the greatest entropy among all things of a given size, Bekenstein and Hawking showed that the amount of entropy they contain is less than what we'd naïvely guess.

That entropy is proportional to surface area is not merely a curious distinction between black holes and Tupperware, about which we can take note and swiftly move on. We've seen that black holes set a limit to the amount of entropy that, even in principle, can be crammed into a region of space: take a black hole whose size precisely equals that of the region in question, figure out how much entropy the black hole has, and that *is* the absolute limit on the amount of entropy the region of space can contain. Since this entropy, as the works of Bekenstein and Hawking showed, is proportional to the black hole's surface area—which equals the surface area of the region, since we chose them to have the same size—we conclude that the maximal entropy any given region of space can contain is proportional to the region's surface area.[3]

The discrepancy between this conclusion and that found from thinking about air trapped in Tupperware (where we found the amount of entropy to be proportional to the Tupperware's *volume*, not its surface area) is easy to pinpoint: Since we assumed the air was uniformly spread, the Tupperware reasoning ignored gravity; remember, when gravity matters, things clump. To ignore gravity is fine when densities are low, but when you are considering large entropy, densities are high, gravity matters, and the Tupperware reasoning is no longer valid. Instead, such extreme conditions require the gravity-based calculations of Bekenstein and